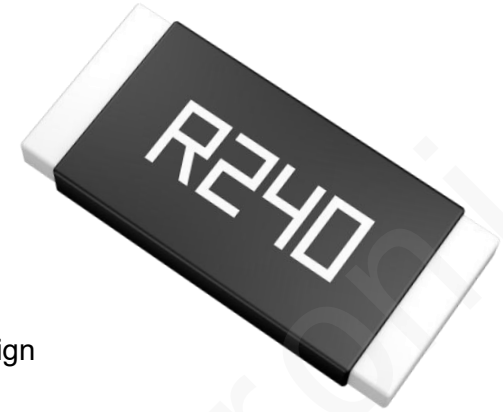
HoJLR2512 Encapsulated Alloy Resistor Series Datasheet

■ Product Features



① Alloy Chip
② High Precision
③ High Reliability
④ Excellent Solderability
⑤ High Overload Capability
⑥ Molded Packaging Process
⑦ RoHS Compliant and Halogen-free
⑧ Wide Operating Temperature Range, Non-inductive Design

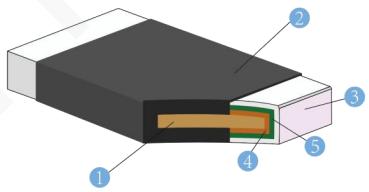
■ Applications

① Inverter Drives
② Servo Drive Systems
③ Power Supply Modules
④ Automation Control Systems
⑤ Industrial Instrumentation & Equipment

■ Electrical Parameters

Part No.	Power	Resistance Range	TCR	Operating Temperature Range	Resistance Tolerance	Max Rated Current	Max Overload Current
2512	2W 3W	0R	/	-55℃~+170℃	±0.5% ±1% ±5%	-	/
		1mR、1.5mR	±250 ppm/℃			54.77A	122.47A
		2mR ~ 510mR	±50 ppm/℃			38.72A	86.60A
2512（S）	3W	0R	/			/	/
		0.5mR	±350ppm/℃			77.45A	173.20A
		1mR、1.5mR	±150pm/℃			54.77A	122.47A
		2mR ~ 4mR	±50ppm/℃			38.72A	86.60A

■ Product Structure



No.	Part Name
1	Alloy Material
2	Molded Compound Layer
3	Tin Layer
4	Copper Layer
5	Nickel Layer

High-tech enterprises IATF16949 ISO9001 RoHS REACH

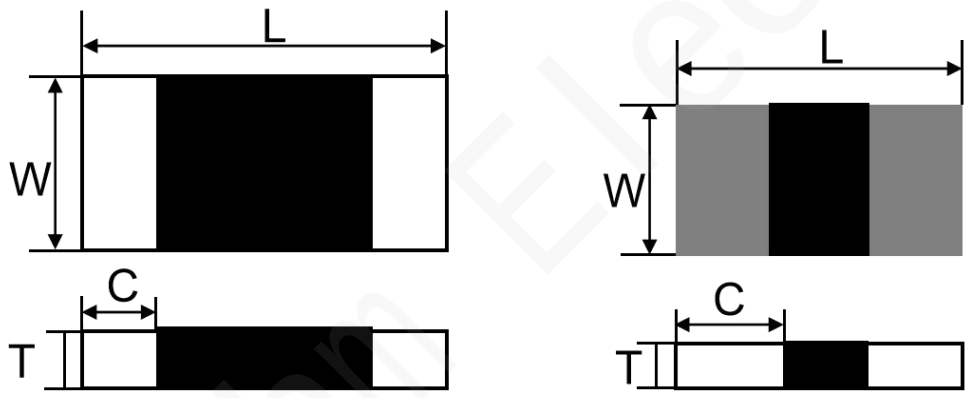
■ Product Selection

Selection Example：HoJLR2512-3W-10mR-1%	Encapsulated Alloy Resistor 2512 3W 10mR 1% Standard Small Electrode

Ho	JLR	2512	3W	10mR	1%	
↓	↓	↓	↓	↓	↓	↓
Manufacturer	**Product Category**	**Package**	**Rated Power**	**Resistance**	**Tolerance**	**Pad**
Ho	JLR	2512	2W 3W	0R-510mR	±0.5% ±1% ±5%	S=Large Electrode (Standard Small Electrode, no letter by default)

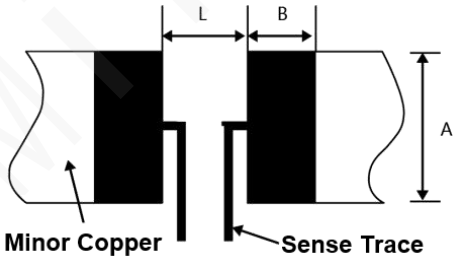
Specific parameters, please refer to the details on the next page.

■ Product Dimensions (Unit: mm)



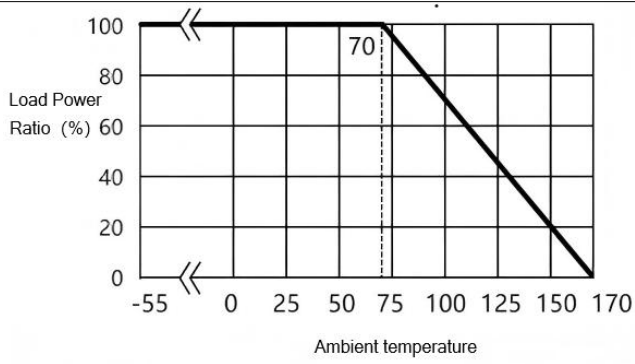
Package	Resistance	L	W	C	T
2512	0R~510mR	6.4±0.2	3.2±0.2	0.9±0.25	0.9±0.2
2512（S）	0R~4mR	6.4±0.2	3.2±0.2	2.1±0.25	0.9±0.2

■ Recommended Pad Dimensions

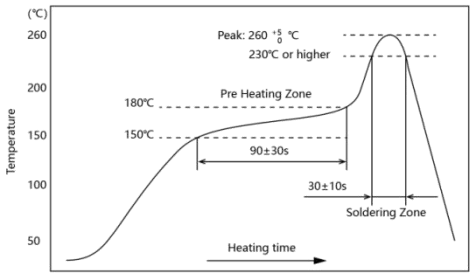
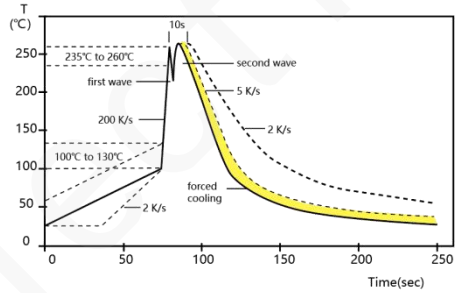


Resistance	A	L	B
0R~510mR	4.0±0.1	4.1±0.1	2.1±0.1
（S）0R~4mR	4.0±0.1	1.3±0.1	3.1±0.1

■ Power Derating Curve



■ Recommended Soldering Parameters

Reflow Soldering Profile	Wave Soldering Profile
	

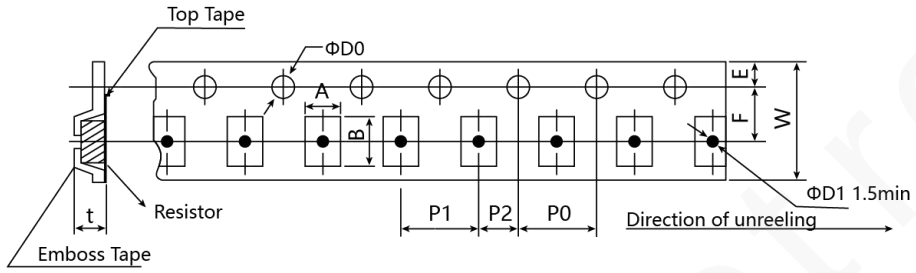
■ Reliability Test

Test Item	Standard	Test Condition	Test Limit Value
Temperature Coefficient of Resistance	IEC60115-1-4.8 JIS-C5201-4.8	+25℃ ~ +125℃	As specified
Load Life	IEC60115-1-4.25.1 JIS-C5201-4.25.1	1000 hours under rated power at 70℃, 1.5h ON / 0.5h OFF cycle	< ±1%
Short-time Overload	IEC60115-1-4.13 JIS-C5201-4.13	5 times rated power, 5 seconds	< ±1%
Humidity Test (No Load)	IEC60115-1-4.24.2.1a) JIS-C5201-4.24.2.1a)	85℃, 85% relative humidity, 1000 hours, with 10% rated power applied	< ±1%
Temperature Cycling	IEC60115-1-4.19 JIS-C5201-4.19	1000 cycles of - 55℃ ~ +170℃, 15 minutes dwell at each extreme condition	< ±1%
Solder Heat Resistance	IEC60115-1-4.18 JIS-C5201-4.18	260±5℃, 10±1 seconds	< ±0.5%
Solderability	IEC60115-1-4.17 JIS-C5201-4.17	245±5℃, 2±0.5 seconds	At least 95% of the electrode surface area shall be covered with new solder
High Temperature Exposure	IEC60115-1-4.23.2 JIS-C5201-4.23.2	170℃, 1000 hours	< ±1%
Low Temperature Storage	IEC60115-1-4.23.4 JIS-C5201-4.23.4	-55℃, 1000 hours	< ±1%
Substrate Bending	IEC60115-1-4.33 JIS-C5201-4.33	Bending deflection: 2mm	< ±0.5%
Insulation Resistance	IEC60115-1-4.6 JIS-C5201-4.6	100 VDC for 1 minute	> 100MΩ

■ Rated Current Calculation Formula

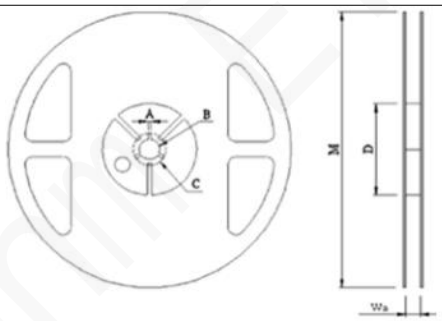
Rated Current Calculation Formula			
$I=\sqrt{P/R}$	I	P	R
	Rated Current (A)	Rated Power (W)	Resistance (Ω)

■ TapeSpecifications(mm)



A	B	W	F	E	P1	P2	P0	D0	T
3.6±0.2	6.9±0.2	12±0.2	5.5±0.05	1.75±0.1	4.0±0.1	2±0.1	4±0.05	Φ1.5+0.1	1.2±0.15

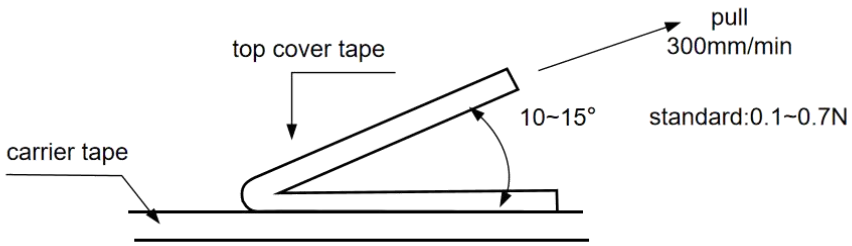
■ ReelSpecifications（mm）



A	B	C	D	M	W	Quantity
2±0.5	13.5±0.5	21±0.5	60±1	178±2	13.8±0.5	4000pcs

■ Peeling Strength of Top Cover Tape

Peeltest:peelspeed 300mm/min;peelforce 0.1N-0.7N



■ Product Usage Recommendations

- ◆ During product usage, protect the surface to prevent defects such as bumps and scratches.
- ◆ Avoid subjecting the product to mechanical stress during installation and operation.
- ◆ The long-term operating power of the product shall be less than or equal to the rated power to prevent resistance drift caused by long-term overload.
- ◆ When the product is used under high temperature or poor heat dissipation conditions, derating shall be applied referring to the power derating curve.

■ Storage Instructions

- ◆ The product storage ambient temperature shall be 5~35℃, humidity ＜65%RH, and humidity shall be kept as low as possible.
- ◆ The product shall be stored in a clean, dry environment free of harmful gases.
- ◆ Do not take the product out of the tape-and-reel packaging before use.
- ◆ Under the above storage conditions, the product shelf life is 1 year.
- ◆ For products stored more than 1 year, check for surface oxidation and perform soldability test.

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■ Revision History:

Version No.	Revision Date	Revision Content	Reason for Revision	Changed By	Reviewed By
Ho-A1	2021-10-22	Changed package dimensions	Product Standardization	Huang Yongkang	Leng Wenyi
Ho-A2	2021-11-22	Updated short-time overload test multiplier	Product Optimization	Huang Zhongliang	Leng Wenyi
Ho-A3	2022-03-28	Discontinued large solder pad specification	Product Standardization	Huang Yongkang	Leng Wenyi
Ho-A4	2024-06-01	Updated reliability test item details	Product Optimization	Huang Yongkang	Leng Wenyi
Ho-A5	2024-07-29	Added 0.5~4mR resistance range for large pad version	Product Upgrade Layout Unification	Huang Yongkang	Leng Wenyi
		Standardized resistance values, dimensions and tolerances			Leng Wenyi
		Adjusted operating temperature range and updated datasheet layout			Leng Wenyi
Ho-A6	2024-11-14	Updated datasheet layout	Layout Unification	Huang Yongkang	Leng Wenyi
Ho-A7	2024-12-13	Added maximum overload current and application scope to electrical parameter section	Content Update	Huang Yongkang	Leng Wenyi
Ho-A8	2025-09-05	Detailed update of TCR and expanded available resistance range	Product Upgrade	Huang Yongkang	Leng Wenyi

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